```json
{
"name": "WF4_Sync_Engine_Event_Driven",
"nodes": [
{
"parameters": {
"path": "gcal-sync-trigger",
"httpMethod": "POST",
"responseMode": "lastNode",
"options": {}
},
"name": "Webhook",
"type": "n8n-nodes-base.webhook",
"typeVersion": 2.1,
"position": [0, 0],
"webhookId": "gcal-sync-trigger",
"id": "wf4_webhook_001"
},
{
"parameters": {
"jsCode": "// Validate webhook payload from Google Apps Script\nconst input =
$input.first().json;\n\n// Verify source\nif (input.source !== 'google_apps_script') {\n
throw new Error('Invalid source: ' + input.source);\n}\n\n// Extract event data\nconst event
= input.event || {};\n\nreturn [{\n  json: {\n    calendar_id: input.calendar_id,\n
event_id: event.id,\n    event_title: event.title,\n    event_start: event.start,\n
event_end: event.end,\n    event_status: event.status,\n    event_description:
event.description,\n    last_updated: event.lastUpdated,\n    html_link: event.htmlLink,\n
sync_type: input.sync_type,\n    received_at: new Date().toISOString()\n  }\n}];",
"options": {}
},
"name": "Validate Webhook",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [250, 0],
"id": "wf4_validate_002"
},
{
"parameters": {
"conditions": {
"string": [
{
"value": "cancelled",
"operation": "equals",
"property": "={{ $json.event_status }}"
}
]
}
},
"name": "Is Event Cancelled?",
"type": "n8n-nodes-base.if",
"typeVersion": 2.3,
"position": [500, 0],
"id": "wf4_if_cancelled_003"
},
{
"parameters": {
"operation": "executeQuery",
"query": "-- Find booking by GCal event ID and cancel it\nUPDATE bookings \nSET status =
```

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'cancelled', \n    cancelled_at = NOW(), \n    cancellation_reason = 'Event deleted/cancelled
in Google Calendar',\n    gcal_synced_at = NOW()\nWHERE gcal_event_id = $1::text \nRETURNING
booking_id, status, customer_id;",
"options": {
"queryReplacement": "={{ $json.event_id }}"
}
},
"name": "Cancel DB Booking",
"type": "n8n-nodes-base.postgres",
"typeVersion": 2.6,
"position": [750, -100],
"credentials": {
"postgres": {
"id": "SFNQsmuu4zirZAnP",
"name": "Postgres account"
}
},
"continueOnFail": true,
"id": "wf4_cancel_004"
},
{
"parameters": {
"operation": "executeQuery",
"query": "-- Find or update booking by event ID\n-- First try to find existing booking by
event ID\nWITH existing AS (\n  SELECT booking_id FROM bookings \n  WHERE gcal_event_id =
$1::text \n  LIMIT 1\n),\n-- If not found, try to find by time match\nmatched AS (\n  SELECT
booking_id FROM bookings \n  WHERE DATE(start_time) = DATE($2::timestamp)\n    AND
EXTRACT(HOUR FROM start_time) = EXTRACT(HOUR FROM $2::timestamp)\n    AND provider_id =
(SELECT provider_id FROM services WHERE service_id = 1) -- Default provider\n  LIMIT
1\n)\nSELECT COALESCE(\n  (SELECT booking_id FROM existing),\n  (SELECT booking_id FROM
matched)\n) as booking_id;",
"options": {
"queryReplacement": "={{ $json.event_id }}, {{ $json.event_start }}"
}
},
"name": "Find Matching Booking",
"type": "n8n-nodes-base.postgres",
"typeVersion": 2.6,
"position": [750, 100],
"credentials": {
```

```
"postgres": {
"id": "SFNQsmuu4zirZAnP",
"name": "Postgres account"
}
},
"id": "wf4_find_005"
},
{
"parameters": {
"conditions": {
"string": [
{
"value": "",
"operation": "notEmpty",
"property": "={{ $json.booking_id }}"
}
]
```

}
},
"name": "Booking Found?",
"type": "n8n-nodes-base.if",
"typeVersion": 2.3,
"position": [1000, 100],
"id": "wf4_if_found_006"
},
{
"parameters": {
"operation": "executeQuery",
"query": "-- Update existing booking with GCal event ID and sync timestamp\nUPDATE bookings
\nSET gcal_event_id = $1::text,\n   gcal_synced_at = NOW(),\n   status = CASE WHEN status
!= 'cancelled' THEN 'confirmed' ELSE status END\nWHERE booking_id = $2::int\nRETURNING
booking_id, gcal_event_id;",
"options": {
"queryReplacement": "={{ $('Validate Webhook').first().json.event_id }}, {{ $json.booking_id
}}"
}
},
"name": "Update Booking Sync",
"type": "n8n-nodes-base.postgres",
"typeVersion": 2.6,
"position": [1250, 0],
"credentials": {
"postgres": {
"id": "SFNQsmuu4zirZAnP",
"name": "Postgres account"
}
},
"id": "wf4_update_007"
},
{
"parameters": {
"jsCode": "// No matching booking found - this is an orphan event\n// Log it for manual
review\nconst event = $('Validate Webhook').first().json;\n\nreturn [{\n json: {\n
action: 'orphan_event_detected',\n   event_id: event.event_id,\n   event_title:
event.event_title,\n   event_start: event.event_start,\n   message: 'Event exists in GCal
but not in DB - may need manual review',\n   requires_action: true\n }\n}];",
"options": {}
},
"name": "Handle Orphan Event",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [1250, 200],
"id": "wf4_orphan_008"
},
{
"parameters": {
"operation": "executeQuery",
"query": "-- Log sync operation\nINSERT INTO system_logs (workflow_name, execution_id, level,
message, details) \nVALUES ($1::text, $2::text, 'INFO', $3::text, $4::jsonb) \nRETURNING
id;",
"options": {
"queryReplacement": "={{ $node[\"Validate Webhook\"].json[\"sync_type\"] || 'gcal_sync' }},
{{ $execution.id }}, GCal sync processed, {{ JSON.stringify($input.first().json) }}"
}
},

"name": "Log Sync Operation",
"type": "n8n-nodes-base.postgres",
"typeVersion": 2.6,
"position": [1500, 100],
"credentials": {
"postgres": {
"id": "SFNQsmuu4zirZAnP",
"name": "Postgres account"
}
},
"continueOnFail": true,
"id": "wf4_log_009"
},
{
"parameters": {
"jsCode": "// Standard Contract Output\nconst input = $input.first().json;\nconst webhookData = $('Validate Webhook').first().json;\n\nreturn [{\n json: {\n   success: true,\n error_code: null,\n   error_message: null,\n   data: {\n   synced: true,\n calendar_id: webhookData.calendar_id,\n   event_id: webhookData.event_id,\n action_taken: input.action || 'sync_completed',\n   booking_id: input.booking_id || null,\n   requires_manual_review: input.requires_action || false\n   },\n   _meta: {\n source: 'WF4_SYNC_ENGINE_EVENT_DRIVEN',\n   timestamp: new Date().toISOString(),\n version: '2.0.0'\n   }\n }\n}];",

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"options": {}
},
"name": "Standard Contract Output",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [1750, 100],
"id": "wf4_sco_010"
},
{
"parameters": {
"jsCode": "// Handle cancelled event result\nconst cancelled = $input.first().json;\n\nreturn [{\n  json: {\n    action: 'booking_cancelled',\n    booking_id: cancelled.booking_id,\n status: cancelled.status,\n    customer_id: cancelled.customer_id,\n    message: 'Booking cancelled due to GCal event deletion'\n  }\n}];",
"options": {}
},
"name": "Cancel Result",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [1000, -100],
"id": "wf4_cancel_result_011"
}
],
"connections": {
"Webhook": {
"main": [
[
{
"node": "Validate Webhook",
"type": "main",
"index": 0
}
]
]
```

```
},
"Validate Webhook": {
"main": [
[
{
"node": "Is Event Cancelled?",
"type": "main",
"index": 0
}
]
]
},
"Is Event Cancelled?": {
"main": [
[
{
"node": "Cancel DB Booking",
"type": "main",
"index": 0
}
],
[
{
"node": "Find Matching Booking",
"type": "main",
"index": 0
}
]
]
},
"Cancel DB Booking": {
"main": [
[
{
"node": "Cancel Result",
"type": "main",
"index": 0
}
]
]
},
"Cancel Result": {
"main": [
[
{
"node": "Log Sync Operation",
"type": "main",
"index": 0
}
```



```
]
]
},
"Find Matching Booking": {
"main": [
[
{
"node": "Booking Found?",
```

"type": "main",
"index": 0
}
]
]
},
"Booking Found?": {
"main": [
[
{
"node": "Update Booking Sync",
"type": "main",
"index": 0
}
],
[
{
"node": "Handle Orphan Event",
"type": "main",
"index": 0
}
]
]
},
"Update Booking Sync": {
"main": [
[
{
"node": "Log Sync Operation",
"type": "main",
"index": 0
}
]
]
},
"Handle Orphan Event": {
"main": [
[
{
"node": "Log Sync Operation",
"type": "main",
"index": 0
}
]
]
},
"Log Sync Operation": {
"main": [
[
{
"node": "Standard Contract Output",
"type": "main",
"index": 0
}
]
]
}
},

"settings": {
"executionOrder": "v1"
},
"active": false,
"versionId": "wf4-sync-event-driven-v2.0.0"
}